

VL 401: RTL Design and Verification

Course Code: VL 401

Course Name: RTL Design and Verification

L-T-P-C: 2-0-2-3

Prerequisites: Computer Organization and Processor Architecture Design (EE 326) or equivalent.

Students intended for: Core course for B. Tech in Microelectronics and VLSI (3rd Year).

Elective or Core: Core

Mutual Exclusion: -

Approval: 59th BoA

Course Contents

- **Introduction to RTL Design:** The concept of RTL abstraction and its role in modern VLSI design; Design methodologies and tool flows used in industry; Digital VLSI Design Flow; High Level Design Representation; Transformations for High Level Synthesis. [2 Lectures]
- **Hardware-synthesizable RTL Design:** Levels of abstractions in RTL; Pipelining of a design; Hardware-efficient arithmetic implementations; Hardware-software partitioning; Parallel processing; Dataflow-based hardware design; Hardware accelerators; Memory architectures for RTL-based designs. [5 Lectures]
- **Fundamentals of HLS, Logic Optimization and Synthesis:** Fundamental of High-Level Synthesis (HLS) including scheduling, allocation, and binding; Various scheduling algorithms and their impact on design performance; Efficient resource allocation and binding strategies; Logic optimization; Boolean logic minimization and optimization techniques; Two level Boolean Logic Synthesis; Heuristic Minimization of Two-Level Circuits; ESPRESSO logic minimizer; Finite State Machine Synthesis; Multilevel Implementation. [5 Lectures]
- **Temporal Logic:** Formal methods for verification using temporal logic are introduced. Fundamental operators and their applications in model checking are discussed. Introduction to formal methods for verification; Temporal Logic: Introduction and Basic Operators; Syntax and Semantics of CTL; Equivalence between CTL Formulas. [3 Lectures]
- **Foundations of Discrete Mathematics in Digital Logic:** Role of discrete mathematics in hardware verification; Graph Theory Basics for Logic Representation; Trees, directed acyclic graphs (DAGs), and their importance in logic design; Adjacency matrices and list representations. [3 Lectures]
- **Binary Decision Diagram:** Binary Decision Diagrams (BDDs) representation and manipulation of Boolean functions; Construction techniques in formal verification; Binary Decision Diagram: Introduction and construction; Ordered Binary Decision Diagram; Operations on Ordered Binary Decision Diagram; Ordered Binary Decision Diagram for Sequential Circuits; Reduced Ordered Binary Decision Diagram (ROBDD). [3 Lectures]
- **Verification Techniques:** Fundamental and advanced verification techniques; Model checking and symbolic model checking techniques; Advance Verification Techniques; Model Checking; Symbolic Model Checking. [5 Lectures]

Laboratory/practical Modules:

1. Hands-on experience with RTL coding and synthesis using Verilog/SystemVerilog.
2. Simulation of combinational and sequential circuits using industry-standard tools.
3. Implementation of functional and formal verification techniques.
4. Development of testbenches and assertion-based verification strategies.
5. Setting up a UVM—based verification environment for large-scale designs.

Text books:

1. Lionel Benipg and Harry Foster, **Principles of Verifiable RTL Design: A functional coding style supporting verification processes in Verilog**, Springer, New York, 2001
2. Giovanni De Micheli, **Synthesis and Optimization of Digital Circuits**, McGraw Hill, 2012.

References:

1. Chris Spear, **System Verilog for Verification: A Guide to Learning the Testbench Language Features**, 2nd Edition, Springer, 2012.
2. Sanjay Churiwala and Sapan Garg, **Principles of VLSI RTL Design: A Practical Guide**, Springer, 2011
3. Samir Palnitkar, **Verilog HDL: A Guide to Digital Design and Synthesis**, Prentice Hall